2016 EXAMINATIONS

Lancaster University

Part II

COMPUTING AND COMMUNICATIONS [2.5 Hours]

SCC.312 Languages and Compilation

Candidates are asked to answer THREE questions from FOUR; each question is worth a total of 25 marks.

Use a separate answer book for each question.

Question 1

Consider the following context-free grammar:

$$S \rightarrow aSa \mid bSb \mid c$$

1.a Describe in words the set of valid sentences from this grammar.

[2 marks]

1.b Draw a *non-deterministic* push-down recogniser (PDR) that corresponds directly to the above context-free grammar with the usual three states.

[5 marks]

1.c Working from the set of valid sentences, now create an equivalent *deterministic* PDR.

[10 marks]

1.d Draw a *deterministic finite state recogniser (FSR)* that recognises sentences derived from the following set definition:

$$\{ b^i c^{2j} : i \geq 0, j \geq 1 \}$$

[2 marks]

1.e Write all the production rules in the grammar corresponding to your FSR in part d.

[2 marks]

1.f Give two examples of valid sentences for your grammar and two examples of invalid sentences.

[2 marks]

1.g Now modify your FSR so that it recognizes sentences derived from the modified set definition:

$$\{ (ab)^i c^{2j} : i \geq 0, j \geq 1 \}$$

[2 marks]

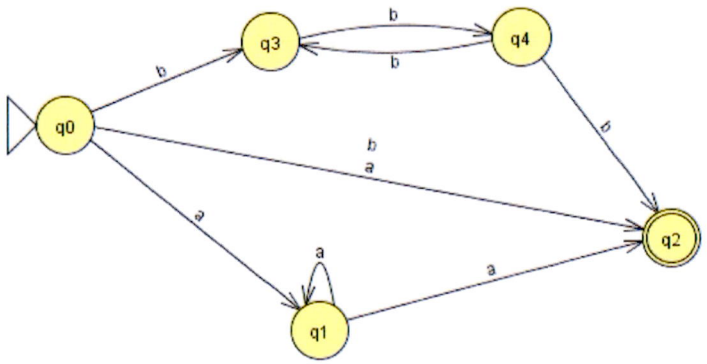
Total 25 marks

Question 2

2.a Define the subset construction algorithm by describing its purpose and listing the main steps in the algorithm.

[6 marks]

2.b Consider the following FSR. Using the subset construction algorithm, generate a new deterministic FSR from it. Make sure that you show the separate steps that you have listed in part a.



[8 marks]

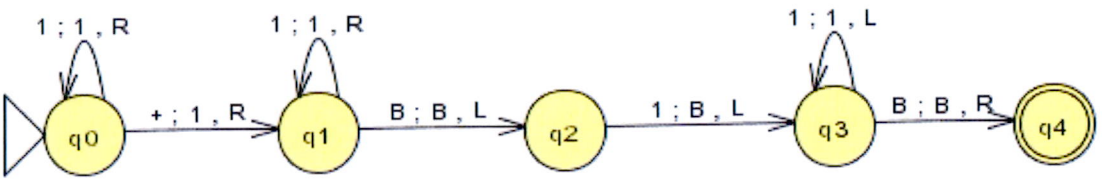
2.c Express the language accepted by your deterministic finite state recognizer as a *regular expression*.

[2 marks]

2.d Type 0 (unrestricted grammars) sit at the top of the Chomsky Hierarchy. Describe the format of the production rules for type 0 grammars and say how they differ from the production rules for type 1 (context sensitive) grammars.

[5 marks]

2.e Consider the Turing Machine below which carries out an unspecified operation on its input tape. Code this Turing Machine ready for use in a Universal Turing Machine as a set of quintuples and show the result in tablular form.



[4 marks]

Total 25 marks

Question 3

Here is a grammar G, with terminals {h, e, l, o}, non-terminals {H, E, L} and H is the distinguished symbol.

H -> h E

E -> e L

L -> l L

L -> o

Build the Action and Goto tables for this grammar. In your answer, you **must** use the technique shown in the course.

Marks will be awarded as follows.

i. Step 1 :	augment the grammar	**[1 mark]**
ii. Step 2 :	build the item sets.	**[15 marks]**
iii. Step 3 :	generate the state transition table	**[3 marks]**
iv. Step 4 :	produce the Action and Goto Tables.	**[6 marks]**
		Total 25 marks

Question 4

4.a Here is an augmented grammar G, with terminals {a, b, c}, non-terminals {T, A, B} and T is the distinguished symbol, along with the Action and Goto tables.

(0) S -> T

(1) T -> A b B

(2) T -> B

(3) A -> a B

(4) B -> c

action	a	b	c	$
0	4	5		
1				acc
2		6		
3	r2	r2	r2	r2
4			5	
5	r4	r4	r4	r4
6			5	
7	r3	r3	r3	r3
8	r1	r1	r1	r1

goto	T	A	B
0	1	2	3
1			
2			
3			
4			7
5			
6			8
7			
8			

Parse the sentence "a c b c". Show all working and the changing state of the parser.

[17 marks]

4.b.

i. What kind of information is stored in the symbol table? Your answer should consider simple variables, arrays, functions and record types.

[6 marks]

ii. Show how the symbol table might be used to reveal the error in the following program fragment:

```
public void one()
{
      int x ;
      x = "47" ;
} // end of method one
```

[2 marks]

Total 25 marks

---End of Paper---